232522: metamonzogranite, Balmoral prospect

(Maitland River Supersuite, Central Pilbara Tectonic Zone, Pilbara Craton)

Location and sampling

Yarraloola (SF 50-6), Pinderi Hills (2255)
MGA Zone 50, 480100E 7637370N

Sampled on 29 November 2018

This sample was collected from the 667.62 – 669.35 m depth interval of drillcore 18ABAD01, drilled in 2018 by Artemis Resources Ltd at their Balmoral prospect (Van Alphen, 2018), with support from the WA Government Exploration Incentive Scheme (EIS). The drillhole is located on Yalleen Station, about 19.5 km south of Zebra Hill, 14.0 km north-northeast of Boolomba Pool, and 5.0 km west of the Hamersley Iron (Rio Tinto) railway.

Geological context

The unit sampled is a metamonzogranite of the Maitland River Supersuite (Hickman, 2021a). The 3006–2982 Ma Maitland River Supersuite forms a 50 km-wide belt of tonalite, granodiorite, and monzogranite intrusions that extends for about 250 km along the coastal plain of the northwest Pilbara region. Granitic rocks of the Maitland River Supersuite intrude numerous older units, including the 3024–3007 Ma Orpheus Supersuite (Hickman, 2021b), and are cross-cut by several younger units, including the 2955–2913 Ma Sisters Supersuite (Hickman, 2021c). A porphyritic granodiorite of the Maitland River Supersuite, exposed as an inlier within lower Fortescue Group rocks, about 48 km to the northeast, yielded an igneous crystallization age of 3006 ± 12 Ma (GSWA 168932, Nelson, 2001).

Petrographic description

The sample is an equigranular, medium-grained metamonzogranite (Fig. 1), consisting of about 40% plagioclase, 40% quartz, 15% K-feldspar, 4–5% biotite, 1% muscovite and trace myrmekite, apatite, zircon and sphalerite. Plagioclase (An_{34}, andesine) is anhedral to subhedral, equant to slightly tabular, up to 3 mm long, and variably sericitized. Quartz is anhedral, irregular, up to 5 mm in size, weakly to moderately strained, and exhibits minor subgraining. K-feldspar is anhedral, irregular, strained, up to 2.5 mm in size, and display incipient to well-developed microcline twinning. Lobes of myrmekite, mostly <0.3 mm in size, are developed at the margins between feldspars. Chlorite-altered biotite forms anhedral to subhedral laths up to 1 mm long, and muscovite forms subhedral laths up to 0.6 mm long, associated with biotite. Apatite forms euhedral, bladed grains up to 0.25 mm in length. The sample contains a single 0.4 mm anhedral grain and rare smaller grains of orange–brown sphalerite. The larger sphalerite grain contains tiny chalcopyrite inclusions. The metamonzogranite has undergone weak solid-state deformation at low to moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and anhedral to euhedral. The crystals are up to 450 µm long, and elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and some zircons appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 5–6 September 2019, using SHRIMP-B. Twelve analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 1.00% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.33% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-two analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Six analyses are >5% discordant. The dates obtained from these six analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 16 analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios and their positions within the crystals.

Group I comprises 13 analyses of 10 zircons (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 3004 ± 4 Ma (MSWD = 2.5).

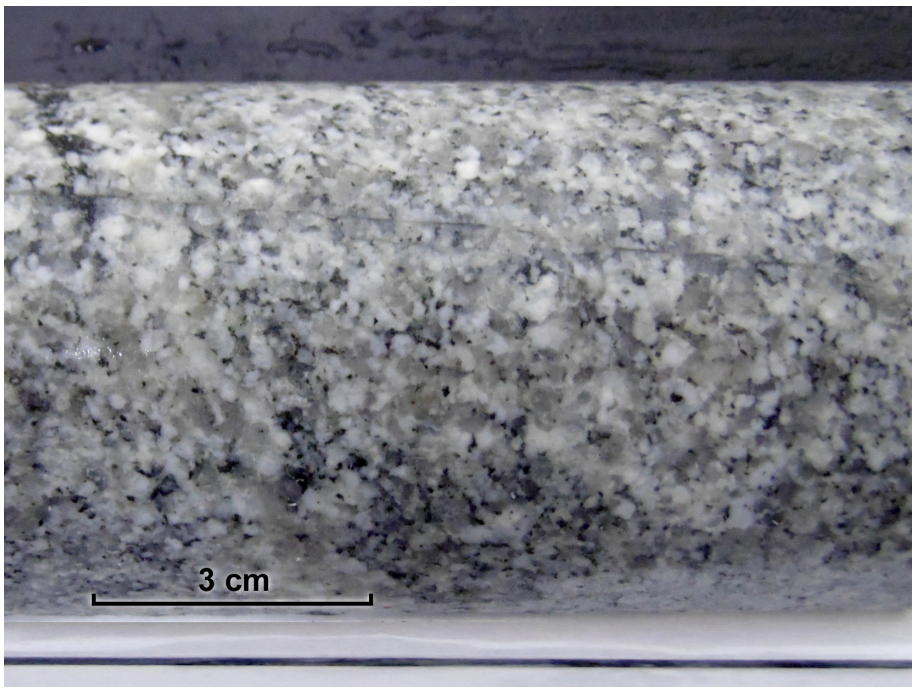

Figure 1. Drillcore image for sample 232522: metamonzogranite, Balmoral prospect

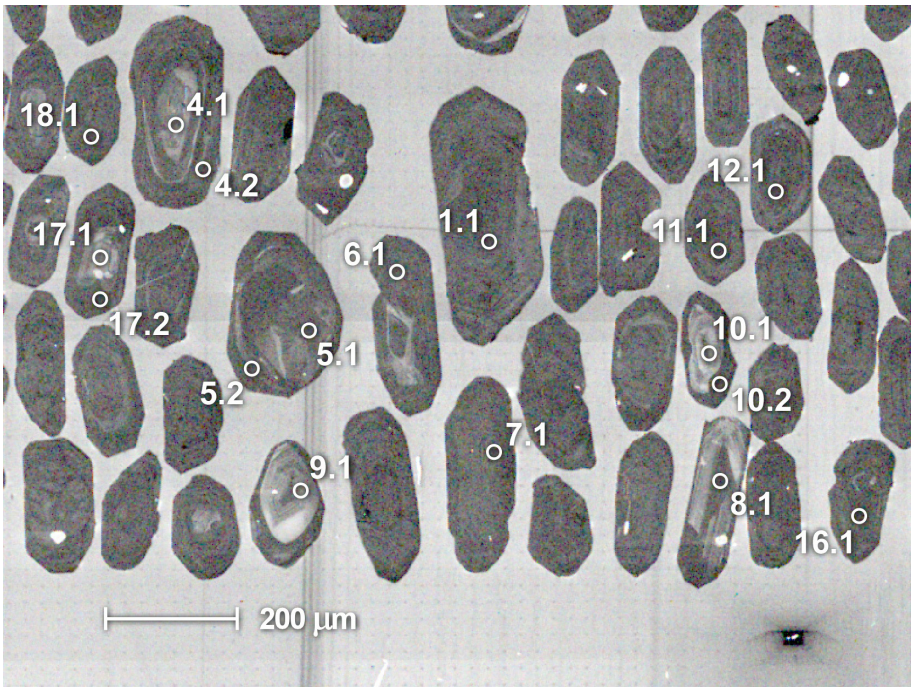

Figure 2. Cathodoluminescence image of representative zircons from sample 232522: metamonzogranite, Balmoral prospect. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 232522: metamonzogranite, Balmoral prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
I	11	5.2	363	57	0.16	0.020	1.721	0.020	0.22203	0.00054	1.721	0.020	0.22185	0.00054	2953	28	2994	4	1.4
I	16	13.1	295	97	0.33	0.050	1.773	0.021	0.22244	0.00060	1.774	0.021	0.22199	0.00061	2882	28	2995	4	3.8
I	13	10.2	361	69	0.19	0.016	1.774	0.021	0.22226	0.00057	1.774	0.021	0.22212	0.00057	2882	28	2996	4	3.8
I	21	17.2	415	83	0.20	0.003	1.775	0.020	0.22280	0.00052	1.775	0.020	0.22278	0.00052	2881	27	3001	4	4.0
I	9	8.1	288	365	1.27	0.000	1.711	0.020	0.22281	0.00063	1.711	0.020	0.22281	0.00063	2967	29	3001	5	1.1
I	3	3.1	306	34	0.11	0.017	1.718	0.020	0.22305	0.00058	1.718	0.020	0.22290	0.00059	2957	28	3002	4	1.5
I	20	17.1	139	68	0.49	0.043	1.774	0.023	0.22331	0.00094	1.774	0.023	0.22293	0.00096	2882	31	3002	7	4.0
I	1	1.1	408	161	0.40	0.032	1.778	0.021	0.22342	0.00054	1.779	0.021	0.22314	0.00055	2876	27	3004	4	4.2
I	12	10.1	94	78	0.83	-0.063	1.719	0.024	0.22289	0.00112	1.718	0.024	0.22344	0.00115	2958	33	3006	8	1.6
I	6	5.1	138	134	0.97	0.044	1.670	0.022	0.22399	0.00086	1.671	0.022	0.22360	0.00088	3024	32	3007	6	-0.6
I	19	16.1	383	109	0.28	0.003	1.725	0.020	0.22398	0.00056	1.725	0.020	0.22396	0.00056	2948	28	3009	4	2.0
I	14	11.1	341	134	0.39	0.006	1.730	0.020	0.22449	0.00055	1.730	0.020	0.22443	0.00055	2940	28	3013	4	2.4
I	15	12.1	506	265	0.52	-0.002	1.710	0.019	0.22450	0.00046	1.710	0.019	0.22451	0.00046	2969	28	3013	3	1.5
X	4	4.1	140	24	0.17	0.013	1.608	0.021	0.25995	0.00194	1.608	0.021	0.25984	0.00194	3117	33	3246	12	4.0
P	10	9.1	48	35	0.71	0.155	1.706	0.028	0.21953	0.00146	1.709	0.028	0.21815	0.00155	2970	39	2967	11	-0.1
P	5	4.2	419	102	0.24	0.000	1.802	0.023	0.22145	0.00053	1.802	0.023	0.22145	0.00053	2846	30	2991	4	4.9
D	17	14.1	740	174	0.23	0.052	2.324	0.042	0.19741	0.00287	2.325	0.042	0.19694	0.00288	2306	35	2801	24	17.7
D	8	7.1	686	188	0.27	0.007	1.880	0.021	0.21231	0.00063	1.880	0.021	0.21225	0.00063	2749	26	2923	5	5.9
D	18	15.1	534	418	0.78	0.047	1.840	0.021	0.21609	0.00044	1.841	0.021	0.21567	0.00045	2797	26	2949	3	5.2
D	7	6.1	312	238	0.76	0.018	1.962	0.041	0.21797	0.00108	1.963	0.041	0.21781	0.00108	2655	46	2965	8	10.5
D	22	18.1	236	239	1.02	0.102	1.828	0.030	0.21983	0.00065	1.830	0.030	0.21892	0.00067	2811	38	2973	5	5.4
D	2	2.1	359	190	0.53	0.010	1.836	0.021	0.22260	0.00058	1.836	0.021	0.22251	0.00059	2803	27	2999	4	6.5

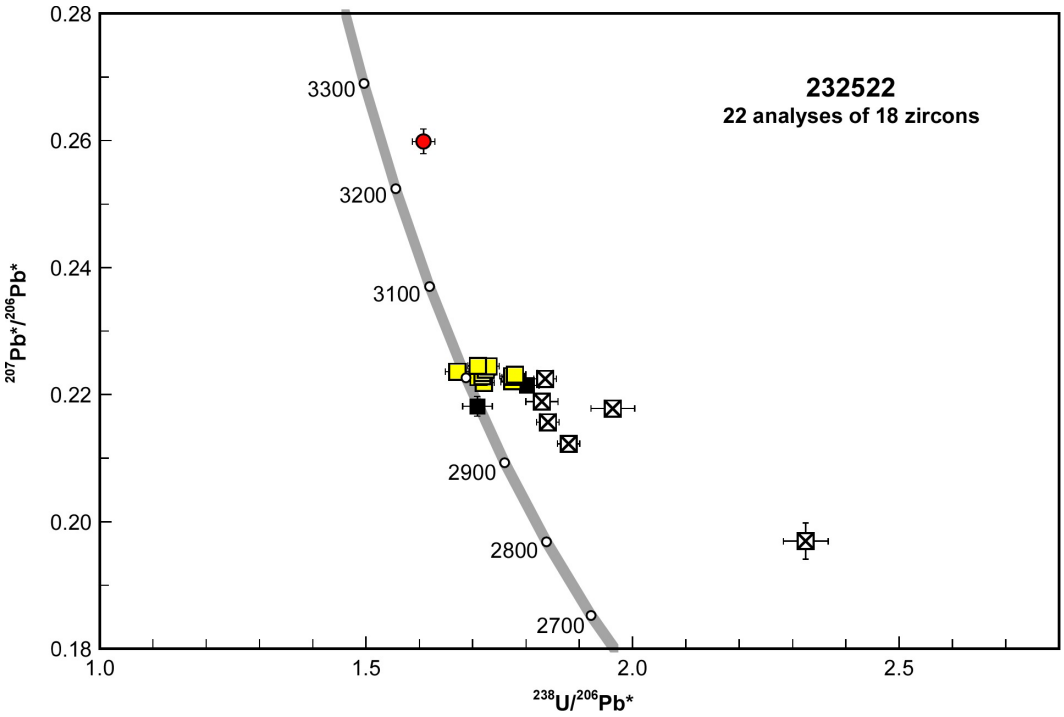

Figure 3. U–Pb analytical data for zircons from sample 232522: metamonzogranite, Balmoral prospect. Yellow squares indicate Group I (magmatic zircons); red circle indicates Group X (xenocrystic zircon core); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%)

Group X comprises one analysis of a zircon core (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 3246 ± 12 Ma (1σ).

Group P comprises two analyses of two zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates (1σ) of 2991 ± 4 and 2967 ± 11 Ma.

The date of 3004 ± 4 Ma for the 13 analyses in Group I is interpreted as the igneous crystallization age of the monzogranite. The date of 3246 ± 12 Ma (1σ) for the single analysis in Group X is interpreted as the age of an inherited zircon. The dates of 2991 and 2967 Ma for the two analyses in Group P are interpreted to reflect loss of radiogenic Pb.

References

Hickman, AH 2021a, Maitland River Supersuite (A-MR-g): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 06 September 2021, <www.dmp.wa.gov.au/ens>.

Hickman, AH 2021b, Orpheus Supersuite (A-OP-xg-o): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 06 September 2021, <www.dmp.wa.gov.au/ens>.

Hickman, AH 2021c, Sisters Supersuite (A-ST-xg-o): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 06 September 2021, <www.dmp.wa.gov.au/ens>.

Nelson, DR 2001, 168932: porphyritic granodiorite, Mardeburra Pool; Geochronology Record 204: Geological Survey of Western Australia, 4p.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Alphen, M 2018, Report on Co-Funded Drilling of Diamond Drillhole 18ABAD01 at Balmoral, West Pilbara (R#17): Geological Survey of Western Australia, Mineral Exploration Report A118065, v. 2, 16p. (unpublished).

Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Howard, HM 2021, 232522: metamonzogranite, Balmoral prospect; Geochronology Record 1817: Geological Survey of Western Australia, 4p.

Data obtained: 6 September 2019

Data released: 12 November 2021